

Edward Zurakowski

Systems & Security Engineer

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Education

University of Wisconsin-Whitewater

Master of Science in Computer Science (Cybersecurity Engineering Emphasis)

December 2026

Bachelor of Science in Computer Science (Minor in Cybersecurity)

December 2025

Skills

- **Languages & Frameworks:** Rust, Java, JavaScript, TypeScript, Python, HTML, Tailwind CSS, SQL, Vue.js, Nuxt
- **Tools & Technologies:** Docker, Git, Woodpecker CI, Linux, REST APIs, Azure, MongoDB, Metasploit
- **Networking & Security:** Multi-tier Clos Topologies (IDF/MOR/TOR), Network Security, TCP/IP, Firewalls, VPNs, Wireshark, Nmap

Experience

TEKsystems (Contracted to Microsoft)

Mount Pleasant, WI

Level 3 Data Center Technician

January 2026 - April 2026

- Operated within Microsoft's Fairwater facility, supporting deployment and maintenance of high-density NVIDIA GB200 NVL72 rack systems.
- Performed fiber optic link validation across multi-tier topologies (T0/T1/T2) from IDF/MOR aggregations to TOR switches.
- Resolved 30+ hardware break-fix tickets, managing physical integrity and power-cycling of compute components and NVIDIA NVLinks.

Noodles & Company

Appleton, WI

Team Member

August 2020 - January 2026

- Trained team members and managed high-volume operational workflows under strict time constraints.

Projects & Academic Research

Torii: Master's Applied Research Project

May 2026 - Present

Zero-Trust eBPF Gateway Capstone

- Architecting a self-hosted gateway bridging Layer 7 verification with kernel-level eBPF packet filtering.
- Building a memory-safe Rust reverse proxy to enforce strict identity verification via OpenID Connect.
- Optimizing HTTP/3 using QUIC connection IDs with eBPF fast-path forwarding for persistent, E2EE tunnels.

AI-Powered Command & Control (C2) Server

January 2026 - May 2026

- Developed a Rust-based payload factory using semaphores for concurrent generation, processing jobs via Redis queues.
- Engineered API endpoints for payload distribution, using PostgreSQL for state persistence and immutable audit logging.
- Built robust background workers with automated filesystem cleanup routines to manage stale build artifacts.

Ember - Cloud-Native Code Snippet Manager

September 2025 - December 2025

- Architectured a full-stack snippet platform featuring complexity analysis, built with Nuxt, TypeScript, BetterAuth, and Tailwind CSS.
- Built an Admin Dashboard with real-time analytics and audit logging to monitor user activity.
- Containerized the application with Docker and established an automated CI/CD pipeline via Woodpecker CI.

HomeServer

2023 - Present

- Manage 50TB of data storage with automated Kopia backups for data resilience and integrity.
- Orchestrate a self-hosted environment using Ubuntu Server, Docker, and Incus for service isolation.
- Hardened network via NetBird VPN, Cloudflare, Caddy, and CrowdSec to control public exposure.

Awards

Boy Scouts of America — *Eagle Scout*

September 2020